Mohammed Nihad

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EDUCATION

University of Virginia **August 2026 - May 2028**
- Bachelor of Science in Computer Engineering
- **Relevant Coursework:** Distributed Systems, Computer Architecture, Database Systems, Linear Systems

City College of New York **August 2024 - May 2026**
- Computer Engineering (5x Hackathon Winner, 3x Dean's List)
- **Relevant Coursework:** Data Structures and Algorithms, Software Engineering and Design, Linear Algebra

SKILLS

- **Languages:** Python, C++, JavaScript, TypeScript, Java, MATLAB, SQL
- **Frameworks:** FastAPI, Spring Boot, React.js, React Native, PyTorch, scikit-learn, Next.js
- **Tools:** Git/ GitHub, AWS, Docker, Redis, PostgreSQL, MongoDB, LangChain, Kubernetes, Grafana

EXPERIENCE

Bloom Energy, Software Engineering Intern **June 2026 – August 2026**
- Built a LangSmith-traced AI agent that autonomously queried Salesforce, OSI PI, AWS Redshift, and internal systems, applied existing diagnostic rules, analyzed historical fault patterns, and generated root-cause and repair recommendations for RMCC escalation cases, solving 45+ tickets and reducing troubleshooting time by 60%.
- Built and shipped a production visual workflow builder using React Flow, FastAPI, and MongoDB, enabling engineers to create reusable automation workflows and reducing script development time by 80%.
- Developed a versioned compiler that converted node-based workflows into validated, Ruff-linted notebooks, configurations, and pipeline files, adopted by a 12-person engineering team.

Rehost, Software Engineering Intern **December 2025 – March 2026**
- Improved 30-day returning users by building an agentic Retention Autopilot that ingests analytics, segments users, and launches approved A/B-tested push + rewards campaigns, cutting campaign setup time by *30%*.
- Reduced AWS Redshift query latency from ~50s to <10s by refactoring nested queries and data-access logic, then exposed the optimized data layer through an MCP server for local developer tooling.

StemKasa, Software Engineering Intern **June 2025 – August 2025**
- Led 3 interns to rebuild a 17-file monolith as a MERN app, implementing component-based architecture, REST APIs, and integration tests while preserving full feature parity.
- Architected and deployed a full CI/CD pipeline using GitHub Actions, automating build, test, and deployment workflows to eliminate manual deployment errors across staging and production environments.
- Engineered an AI tutoring layer with GPT-powered adaptive flashcards and lessons, then integrated secure auth and Stripe subscriptions, driving a 25% engagement increase across 3 schools.

City College of New York, Undergraduate Researcher **June 2025 – December 2025**
- Co-authored research on a self-critiquing LLM pipeline combining expert feedback, distillation, reinforcement learning, and Pareto optimization, improving model accuracy while reducing infrastructure costs by up to 90%.

Universacare, Software Automation Intern **June 2023 – September 2023**
- Built a Python Selenium web scraping workflow to extract daily Home Health Aide data, automating manual work and improving productivity by 11%.
- Developed 15 reusable React components and integrated a role-aware online application form into the production website, improving UI consistency/access control and saving 2+ hours per submission.

PROJECTS

Squeeze (Morgan Stanley 2026 Hackathon Winner) FastAPI | LangChain | LangGraph | Next.js | Supabase
- Built a scalable event-management platform with FastAPI/Next.js backend services, tool orchestration, and AWS Bedrock-powered automation across 12 integrated workflows.
- Project acquired by hackathon partner Lemontree (500K+ accounts) for continued product development.

RhetoriQ (Source Code) Kafka | Flink | Kubernetes | Neo4j | pgvector | LangChain
- Built a low-latency narrative-analysis pipeline using Kafka/Flink, Neo4j graph traversal, pgvector semantic search, Redis caching, and Kubernetes microservices to process 100K+ daily documents.
- Designed a multi-agent orchestration system of 7 specialized AI agents, spanning planning, search, investigation, and synthesis that autonomously traces the origin and spread of political news.
- Developed an agentic LangChain investigation pipeline leveraging HuggingFace NLP models for semantic similarity detection, with Prometheus/Grafana observability for 15+ microservices.